

New features in Predici

Version „Maxwell“ (2024/25)

Major extensions

- ▶ **Coming soon:**
 - ▶ General structures for vectors, matrices, tensors and global variables
 - ▶ New script extensions with loops, object-style commands, shortcuts, array access
- ▶ **Generally:** New version developed in the new Embarcadero **Athens compiler** (2024)
- ▶ **Component windows** smaller and better organized
- ▶ **New Multi-Steps:** New version of the most common steps that can automatically be used for different sites or reactors
 - ▶ supported by GUI and “**generic parameters**”: GP_xxx
- ▶ **Extended heat balance:** Heat exchanger, more options
- ▶ **Shooting control:**
 - ▶ Integrated iteration and parameter tuning
 - ▶ Standard setups accessible by OLE
 - ▶ **HXSpecificationType:** e_spec_jtout=1, e_spec_tr=2, e_spec_jtin=3
 - ▶ **HXTuningType:** e_tune_mp=1, e_tune_hxin=2, e_tune_qadd=3, e_tune_mass=4
- ▶ **OLE/COM:** Number of new commands to control the model

Recipes

► Settings for integration control

☐ use end time: [s]

Exit stream [l/s]	Control stream	Geom. vol. [l]
controlstream 0.00000e+00	Automatic Automatic	1.00e+00

► Tube reactors

Basic definition Additional info Process Control

Unit system Parameter Tube settings

R1

Recipe tube settings overwrite reactor infos

Length m

Diameter m

☐ counter current cooling (heat exch.)

Special options

☐ use prepared parameter list for getkp in scripts

☒ use tube settings from recipes (instead of reactor settings)

► Unit system for each recipe

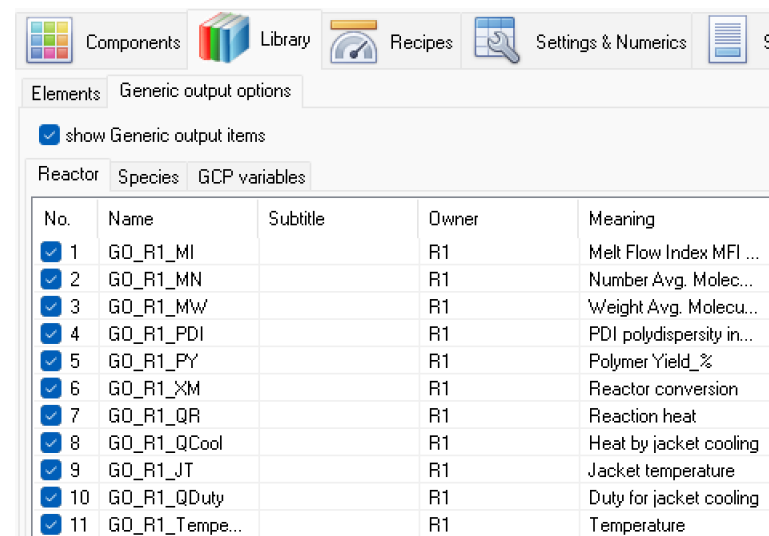
► Feed of polymer in recipes

► Further controls: **shooting** control, pre-schedule

☐ Evaluate script in t<0:

Output extensions

- ▶ **New Generic Outputs:** Extended list with user-defined configuration
 - ▶ groups: reactor, heat balance, species
 - ▶ moments of GPC outputs
- ▶ **Convolution:** Extended option for GPC output variables
- ▶ **Shift** in time axis of outputs by script
- ▶ **Extended online data** (back of Graphic Window)



Generic output options				
☒ show Generic output items				
Reactor				
No.	Name	Subtitle	Owner	Meaning
☒ 1	GO_R1_MI		R1	Melt Flow Index MFI ...
☒ 2	GO_R1_MN		R1	Number Avg. Molec...
☒ 3	GO_R1_MW		R1	Weight Avg. Molecu...
☒ 4	GO_R1_PDI		R1	PDI polydispersity in...
☒ 5	GO_R1_PY		R1	Polymer Yield_%
☒ 6	GO_R1_XM		R1	Reactor conversion
☒ 7	GO_R1_QR		R1	Reaction heat
☒ 8	GO_R1_QCool		R1	Heat by jacket cooling
☒ 9	GO_R1_JT		R1	Jacket temperature
☒ 10	GO_R1_QDuty		R1	Duty for jacket cooling
☒ 11	GO_R1_Tempe...		R1	Temperature

Model extensions

- ▶ **Step extensions:**
 - ▶ PolymerPartition
 - ▶ PDEs/PSDs: FeedProfile, FlowDist, FlowSolve, FluidBalance
- ▶ **Script commands:** few extensions and changes
 - ▶ **getcosum, getholdup getdistholdup** with separator, getcosum("M1;M2")
 - ▶ getmcindex with lumping in reactor

Script editor

- ▶ Direct definition of output
- ▶ Project variables directly accessible
- ▶ Fewer warnings for project name, e.g. in getco

The screenshot displays the Script editor interface. At the top, a 'Name' field contains 'Test'. Below it, a script editor shows the code `result1 = getco("R1:PreciVol")`. To the right, a 'Filter by string' field is present above a table with tabs: 'Template', 'Functions', 'Species' (highlighted with a red box), 'Distributions', and 'Parameters'. The 'Species' table has columns 'No.', 'Name', and 'Reactor', with two rows: 1 CLCB R1 and 2 E R1. Below this, there are tabs for 'Cape Open compounds' and 'Predici Identifier'. A second table, also with 'No.' and 'Name' columns, lists six items: 2 R1:Mass, 3 R1:Vol, 4 R1:PreciVol, 5 R1:Pressure, and 6 R1:HeatExchangerTemp. This table is also highlighted with a red box. At the bottom left, there are buttons for 'Distr. generator', 'unit of result1', and a checkbox for 'is "Action script"'. Below these is a 'Syntax okay' status bar and a 'Syntax' button. At the very bottom, navigation buttons include 'Bw', 'Fw', and a 'Def. output' button highlighted with a red box.

Name: Test

```
result1 = getco("R1:PreciVol")
```

Filter by string:

No.	Name	Reactor
1	CLCB	R1
2	E	R1

Cape Open compounds | Predici Identifier

No.	Name
2	R1:Mass
3	R1:Vol
4	R1:PreciVol
5	R1:Pressure
6	R1:HeatExchangerTemp

unit of result1:

☐ is "Action script"

Syntax okay

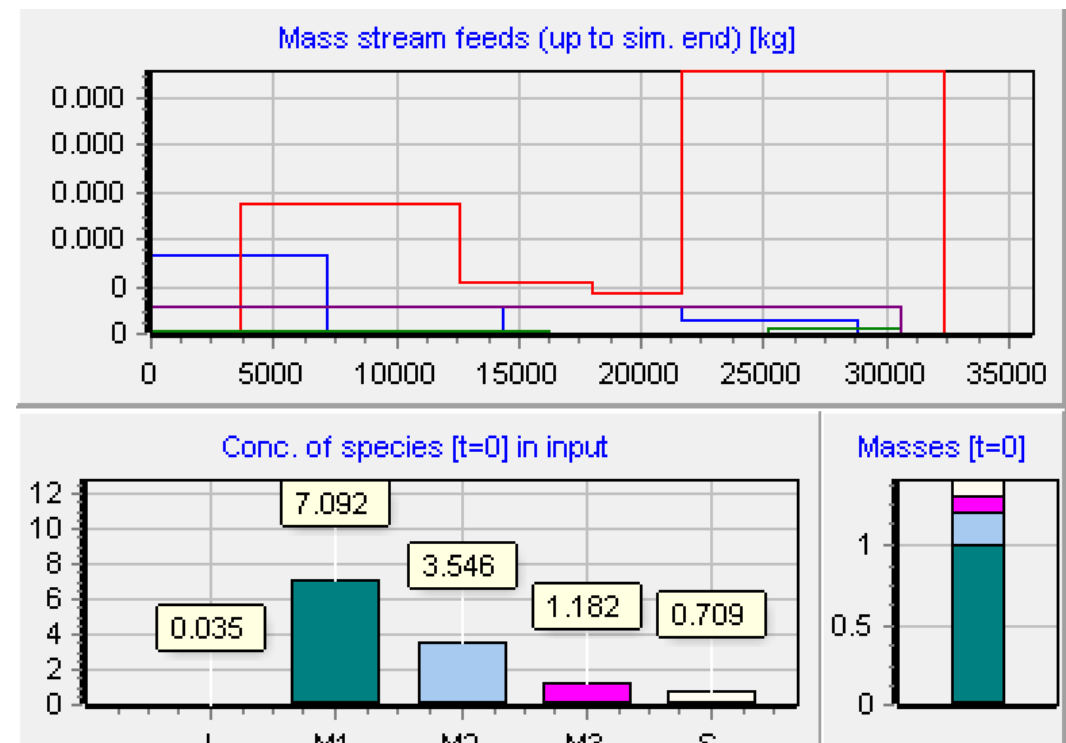
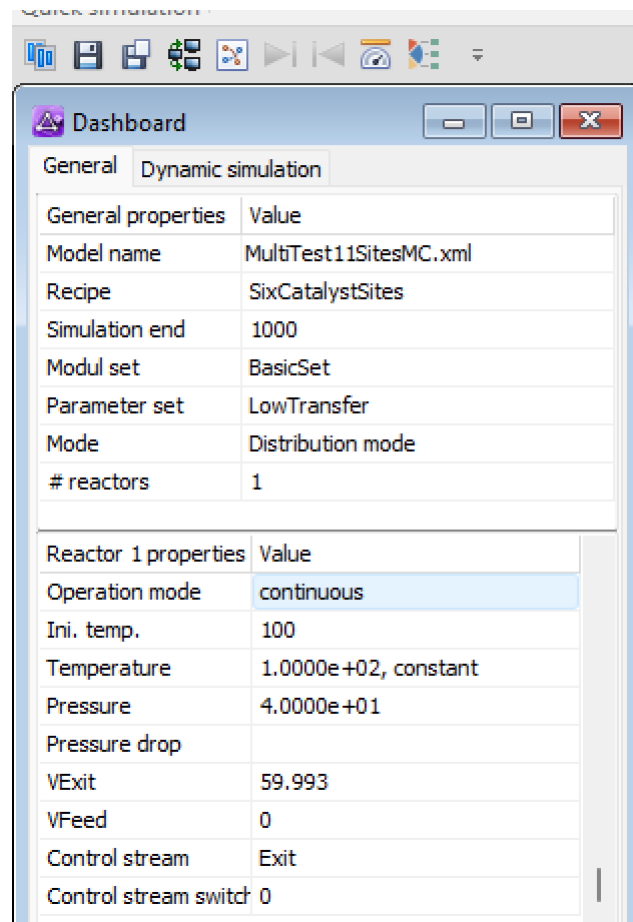
Syntax

Cursor in line: 1

← Bw | Fw → |

Dashboard

- Model and live information in one window



Reactor

- Old settings are finally removed

Name R1

Operation mode: tubular reactor

Temperature settings [°C]: constant, T0 = 200.0000

Pressure settings [bar]: 2000.0000

Tube settings + control | Phases | Heat balance | Volume | Copolymerization | Profile properties | Components

Tube settings (set in recipe)

Length: 5.0000e+02 m

Diameter: 3.0000e-02 m

Mass flow rate: 2.5100e+01 kg/s

☐ based on flow velocity

Velocity: 1 m/s

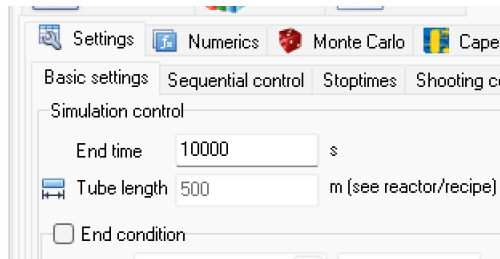
Control stream (see recipe, here additional info)

Switch off after: s

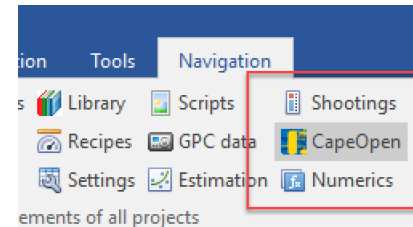
Relative velocity:

Project settings

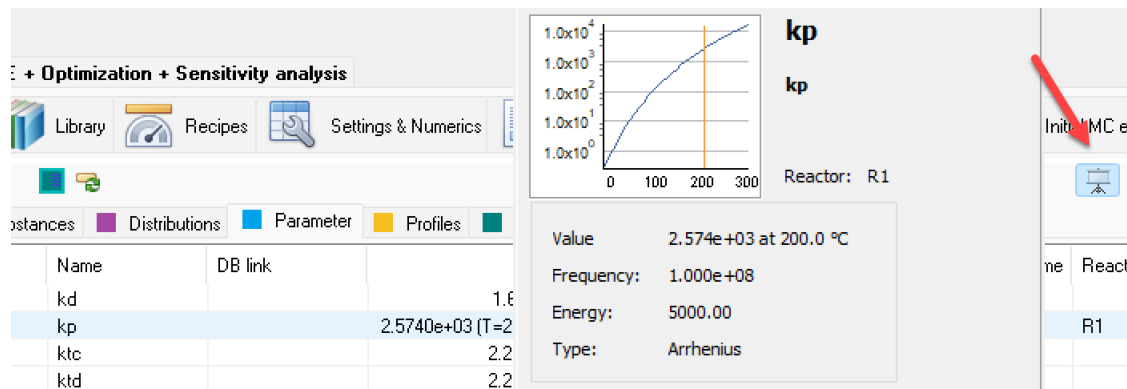
- Project settings for simulation time



- More quick access buttons

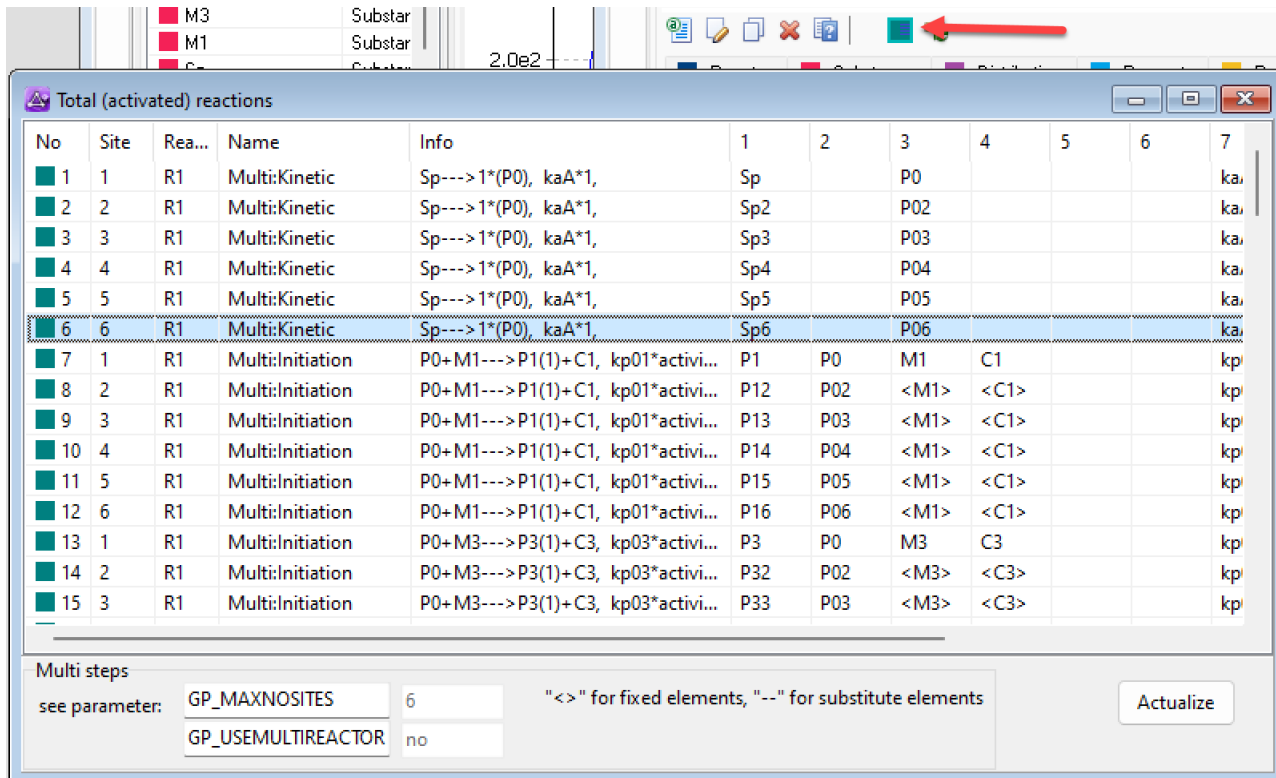


- Project lists: More information available (experimental)



Active-module overview

- Important for multi-site models: shows all active reactions



Substar
Substar
2.0e2

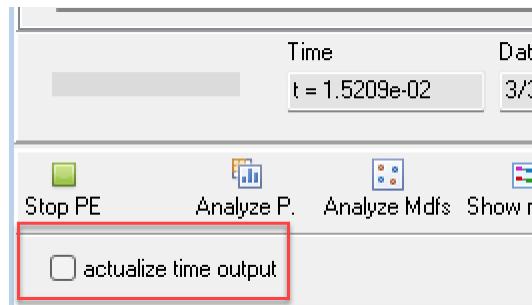
Total (activated) reactions

No	Site	Rea...	Name	Info	1	2	3	4	5	6	7
1	1	R1	Multi:Kinetic	Sp--->1*(P0), kaA*1,	Sp		P0				ka
2	2	R1	Multi:Kinetic	Sp--->1*(P0), kaA*1,	Sp2		P02				ka
3	3	R1	Multi:Kinetic	Sp--->1*(P0), kaA*1,	Sp3		P03				ka
4	4	R1	Multi:Kinetic	Sp--->1*(P0), kaA*1,	Sp4		P04				ka
5	5	R1	Multi:Kinetic	Sp--->1*(P0), kaA*1,	Sp5		P05				ka
6	6	R1	Multi:Kinetic	Sp--->1*(P0), kaA*1,	Sp6		P06				ka
7	1	R1	Multi:Initiation	P0+M1--->P1(1)+C1, kp01*activi...	P1	P0	M1	C1			kp
8	2	R1	Multi:Initiation	P0+M1--->P1(1)+C1, kp01*activi...	P12	P02	<M1>	<C1>			kp
9	3	R1	Multi:Initiation	P0+M1--->P1(1)+C1, kp01*activi...	P13	P03	<M1>	<C1>			kp
10	4	R1	Multi:Initiation	P0+M1--->P1(1)+C1, kp01*activi...	P14	P04	<M1>	<C1>			kp
11	5	R1	Multi:Initiation	P0+M1--->P1(1)+C1, kp01*activi...	P15	P05	<M1>	<C1>			kp
12	6	R1	Multi:Initiation	P0+M1--->P1(1)+C1, kp01*activi...	P16	P06	<M1>	<C1>			kp
13	1	R1	Multi:Initiation	P0+M3--->P3(1)+C3, kp03*activi...	P3	P0	M3	C3			kp
14	2	R1	Multi:Initiation	P0+M3--->P3(1)+C3, kp03*activi...	P32	P02	<M3>	<C3>			kp
15	3	R1	Multi:Initiation	P0+M3--->P3(1)+C3, kp03*activi...	P33	P03	<M3>	<C3>			kp

Multi steps
see parameter: GP_MAXNOSITES 6 " < > " for fixed elements, "--" for substitute elements
GP_USEMULTIREACTOR no Actualize

Parameter estimation

- ▶ **MDF:** Data trimming
- ▶ **OLE** commands
- ▶ Reminder in previous extensions
 - ▶ synopsis tab
 - ▶ batches tab
 - ▶ TRef treatment



New OLE commands (excerpt)

- ▶ HRESULT **CreateSolRecipe**([in] BSTR recipeName, [in] BSTR option);
- ▶ int **SetEnthalpy**([in] BSTR stepType, [in] BSTR reactant, [in] double value);
- ▶ SAFEARRAY(double) **GetDistPoints**([in] BSTR names, [in] int normedOutput, [in] int graphicWeight, [in] int logAxis, [in] int xAsMolWeight);
- ▶ long **FillDistPoints**([in, out] SAFEARRAY(double) points, [in] BSTR names, [in] int normedOutput, [in] int graphicWeight, [in] int logAxis, [in] int xAsMolweight);
- ▶ HRESULT **GetFuncValueFound**([in] BSTR name_of_func, [out] VARIANT* funcFound);
- ▶ HRESULT **SetDistLumping**([in] int onOff);

New OLE commands (excerpt)

- ▶ double **GetReactorPressure**([in] BSTR Name_of_reactor);
- ▶ double **GetDistMoments**([in] BSTR Name_of_distribution, [out] VARIANT* my0, [out] VARIANT* my1, [out] VARIANT* my2, [out] VARIANT* my3, [out] VARIANT* Mn, [out] VARIANT* Mw, [out] VARIANT* Mz, [out] VARIANT* PDI, [out] VARIANT* AMW, [out] VARIANT* mass);
- ▶ double **GetDistMomentsArray**([in] BSTR Name_of_distribution, [out] VARIANT* ptArray);
- ▶ HRESULT **ActivateDetailedIteration**([in] BSTR reactorName, [in] int specType, [in] int tuneType, [in] double specVal, [out] VARIANT* paramName);
- ▶ int **ActivateQBasicIteration**([in] BSTR reactorName, [in] double setPointTemperature, [out] VARIANT* paramName);
- ▶ long **SetHeatExchanger**([in] BSTR reactorName, [in] long useHeatExchanger, [in] double heatTransfer, [in] double area, [in] double heatCapacity, [in] double massFlow, [in] double massHoldup, [in] double initialFeedTemp, [in] long counterCurrent);
- ▶ long **CheckEnthalpy**([in] BSTR reactorName);